A BALLOT-WORD FORMULA FOR Q-ZETA NUMERATORS OF TYPE-B ROOT POSETS

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Abstract. Chapoton's q-Zeta invariant associates a bivariate numerator to a finite ranked poset by weighting weak chains according to rank. We determine this numerator for the standard positive-root poset $P_r = \Phi^+(B_r)$, graded by root height minus one. For every Lie rank $r \geq 1$, we give an explicit positive formula indexed by words of length $2r-2$ whose reversals are ballot words. The proof identifies P_r with a trapezoidal cell poset, adjoins a minimum, and constructs an explicit EL-labelling of the resulting lattice. Maximal-chain descents become the initial positions of the left runs in the corresponding words, and Chapoton's flag expansion then gives the formula. In particular, the numerator is coefficientwise nonnegative, has degree $r-1$ in t, has top t-coefficient $q^{(r-1)(r-2)}$, evaluates to $\binom{2r-2}{r-1}$ at $(q,t) = (1,1)$, and specialises at $q = 1$ to $\sum_{k=0}^{r-1} \binom{r-1}{k}^2 t^k$.

1. Introduction

Positive-root posets connect the combinatorics of finite posets with the structure of crystallographic root systems. Their antichains and order ideals occur naturally in the theory of generalized Catalan numbers and Coxeter arrangements; see, for example, [1, 4]. This makes root posets a natural test class for rank-sensitive refinements of classical order invariants.

Chapoton's q-Zeta invariant refines the ordinary Zeta polynomial of a finite poset by retaining rank information in weak-chain enumeration [3]. The associated rational generating series has a bivariate numerator, but this numerator need not be coefficientwise nonnegative for an arbitrary ranked poset. For the positive-root poset of type B, Chapoton observed that the coefficients of the powers of t appear to be nonnegative q-analogues of type-B Narayana numbers. With the Lie-rank convention used here, the specialization proved below is $[t^k]\mathbb{H}_{P_r,\mathrm{rk}}(1,t) = \binom{r-1}{k}^2$; equivalently, its Narayana parameter is $r-1$. The purpose of this paper is to prove an explicit monomial formula that makes this pattern exact, uniformly in the Lie rank.

This paper treats the full positive-root poset of standard Lie type B_r, where r denotes the Lie rank. The rank convention is part of the statement: throughout, P_r is the full standard positive-root poset and the q-Zeta numerator uses Chapoton's fixed denominator normalization.

Let $P_r = \Phi^+(B_r)$ be ordered by nonnegative simple-root difference and graded by

$$\mathrm{rk}(\alpha) = \mathrm{ht}(\alpha) - 1.$$

The q-Zeta numerator $\mathbb{H}_{P_r,\mathrm{rk}}(q,t)$ is defined precisely in Section 2. Put $m = 2r-2$. Call a word in $\{L, R\}$ a ballot word if every prefix contains at least as many R's as L's, and define

$$\mathcal{B}_r = \{w \in \{L, R\}^m : \text{the reversal of } w \text{ is a ballot word}\}.$$

For $w = w_1 \cdots w_m$, let

$$D(w) = \{j : w_j = L \text{ and } (j = 1 \text{ or } w_{j-1} = R)\};$$

2020 *Mathematics Subject Classification.* Primary 06A07; Secondary 05A15, 17B22.

Key words and phrases. root poset, q-Zeta numerator, flag h-vector, EL-labelling, ballot word, lower-semimodular lattice.

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thus $D(w)$ records the initial positions of the L-runs. For $r = 1$, these definitions use the empty word. The main statement is the following explicit coefficient formula.

Theorem 1.1. *For every integer $r \geq 1$,*

$$\mathbb{H}_{P_r,\mathrm{rk}}(q,t) = \sum_{w\in\mathcal{B}_r} q^{\sum_{j\in D(w)}(j-1)} t^{|D(w)|}.$$

In particular, every coefficient of the q-Zeta numerator of the standard type-B_r positive-root poset is a nonnegative integer.

Example 1.2. The first three numerators are

$$\mathbb{H}_{P_1,\mathrm{rk}}(q,t) = 1, \qquad \mathbb{H}_{P_2,\mathrm{rk}}(q,t) = 1 + t,$$

and

$$\mathbb{H}_{P_3,\mathrm{rk}}(q,t) = 1 + (2 + q + q^2)t + q^2t^2.$$

Thus the formula refines the positivity statement even in small rank: the power of q records the positions at which new left runs begin.

The proof has three structural steps. First, we identify P_r with a trapezoidal cell poset whose order and rank are coordinatewise (Section 3). Second, we show that adjoining a new minimum changes the q-Zeta numerator by the exact substitution $t \mapsto qt$ (Section 4). Third, we equip the augmented cell poset with an EL-labelling by signed endpoint labels. Its maximal chains are precisely the words in $\mathcal{B}_r$, and their descents are the sets $D(w)$ (Sections 5 and 6). We also prove that the augmented poset is a lower-semimodular lattice. For the descent formula we use the explicit EL-labelling directly; no labelling-existence consequence of semimodularity is invoked.

The unrestricted quantifier in Theorem 1.1 is carried entirely by this combinatorial argument. The finite computations described in Section 7.2 are independent consistency checks and are not used to pass from finitely many ranks to all ranks.

Besides positivity, the formula determines the total weight at $(1,1)$, the sharp t-degree, the leading t-coefficient, and the specialization $[t^k]\mathbb{H}_{P_r,\mathrm{rk}}(1,t) = \binom{r-1}{k}^2$ (Corollary 6.2). A further flag interpretation gives a three-rank Euler-characteristic refinement (Corollary 6.4).

2. Preliminaries and conventions

2.1. **Weak chains and the q-Zeta numerator.** Let P be a finite graded poset with a unique maximum and rank function $\rho\colon P \to \{0,\dots,H\}$. For $m \geq 0$, set

$$A_m(P,\rho;q) = \sum_{x_1\leq\cdots\leq x_m} q^{\rho(x_1)+\cdots+\rho(x_m)},$$

where the sum for $m = 0$ consists of the empty chain and equals 1. Chapoton's Lemma 1.1 identifies $A_m(P,\rho;q) = \mathsf{Z}_{P,\rho}([m+1]_q)$ for every $m \geq 1$. For the empty-chain term, Chapoton's Lemma 2.2 gives $\mathsf{Z}_{P,\rho}([1]_q) = \chi(P) = 1$ whenever P has a unique maximum, so the same indexing also holds for $m = 0$.

Definition 2.1. The weak-chain q-Zeta series is

$$\mathcal{F}_{P,\rho}(q,t) = \sum_{m\geq 0} A_m(P,\rho;q)t^m.$$

In the normalization of [3], its numerator is the polynomial $\mathbb{H}_{P,\rho}(q,t)$ determined by

$$\mathcal{F}_{P,\rho}(q,t) = \frac{\mathbb{H}_{P,\rho}(q,t)}{\prod_{a=0}^{H}(1-q^at)}. \tag{1}$$

We use $\mathbb{N} = \{0,1,2,\dots\}$. Thus $\mathbb{H}_{P,\rho} \in \mathbb{N}[q,t]$ means coefficientwise nonnegativity, not merely nonnegativity after specializing q or t.

2.2. **Flag numbers and R-labellings.** Let Q be a bounded graded poset of ranks $0, \dots, H+1$. For $S \subseteq [H] = \{1, \dots, H\}$, let $\alpha_Q(S)$ be the number of chains whose internal set of ranks is exactly S. Its flag h-number is

$$\beta_Q(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} \alpha_Q(T). \tag{2}$$

Write

$$\sigma(S) = \sum_{s \in S} s.$$

An edge labelling of a bounded graded poset by a totally ordered set is an *EL-labelling* if every interval has a unique maximal chain with strictly increasing labels and this chain is lexicographically first among the maximal chains of that interval. An EL-labelling is, in particular, an R-labelling for the usual strict order on labels [2].

The following two standard facts are the only non-elementary poset inputs.

Theorem 2.2 (Chapoton's flag expansion)**.** *For a bounded graded poset Q of ranks $0, \dots, H+1$, equipped with its rank function,*

$$\mathbb{H}_{Q,\mathrm{rk}}(q,t) = \sum_{S \subseteq [H]} \beta_Q(S) q^{\sigma(S)} t^{|S|}. \tag{3}$$

Theorem 2.3 (R-labellings and descents)**.** *If a bounded graded poset Q has an R-labelling, then*

$$\beta_Q(S) = \#\{\textit{maximal chains of } Q \textit{ with descent set } S\} \tag{4}$$

for every set S of internal ranks. In particular, $\beta_Q(S) \geq 0$.

The assertions in Theorems 2.2 and 2.3 are Chapoton's Theorems A.2 and A.1, written in the rank indexing of the present paper: a bounded poset of ranks $0, \dots, H+1$ has internal ranks $[H]$, which is Chapoton's index set $\{1, \dots, H'-1\}$ with $H' = H+1$ [3, Section 5 and Appendix A]. The descent machinery originates in Stanley [5, Theorem 3.1 and Corollary 3.2]; the EL formulation is due to Björner [2].

2.3. **Type-B roots.** In $\mathbb{R}^r$, with standard basis $e_1, \dots, e_r$, we use the simple roots

$$\alpha_i = e_i - e_{i+1} \quad (1 \leq i < r), \qquad \alpha_r = e_r.$$

The positive roots are

$$e_i - e_j \ (i < j), \qquad e_i, \qquad e_i + e_j \ (i < j).$$

For positive roots α, β, the root-poset order is

$$\alpha \leq \beta \quad \iff \quad \beta - \alpha \in \sum_{k=1}^{r} \mathbb{N}\alpha_k.$$

3. A cell model for the type-B root poset

Define the trapezoidal set

$$C_r = \{(i,c) : 1 \leq i \leq r,\ i \leq c \leq 2r - i\}.$$

Give it the order

$$(i,c) \leq (j,d) \quad \iff \quad i \geq j \text{ and } c \leq d. \tag{5}$$

Lemma 3.1. *The map*

$$\begin{aligned} e_i - e_j &\longmapsto (i, j-1), \\ e_i &\longmapsto (i, r), \\ e_i + e_j &\longmapsto (i, 2r-j+1) \end{aligned} \tag{6}$$

is a rank-preserving isomorphism from the positive-root poset $P_r = \Phi^+(B_r)$ to C_r. Under this map,

$$\mathrm{rk}(i, c) = c - i.$$

Proof. For fixed i, the three root families in Equation (6) fill, respectively, columns $i, \dots, r-1$, column r, and columns $r+1, \dots, 2r-i$. The map is therefore a bijection onto C_r.

We verify the order directly in simple-root coordinates. If $v_{i,c}(k)$ is the coefficient of α_k in the root represented by (i, c), then for $c \le r$,

$$v_{i,c}(k) = \begin{cases} 1, & i \le k \le c, \\ 0, & \text{otherwise}, \end{cases}$$

whereas for $c > r$, writing $m = 2r - c + 1$,

$$v_{i,c}(k) = \begin{cases} 0, & k < i, \\ 1, & i \le k < m, \\ 2, & m \le k \le r. \end{cases}$$

The cell condition gives $m \ge i + 1$.

Suppose $i \ge j$ and $c \le d$. If both columns are at most r, the support interval for $v_{i,c}$ is contained in that for $v_{j,d}$. If $c \le r < d$, the latter vector is at least one on the entire interval from j to r. If $r < c \le d$, both the support of the positive entries and the terminal interval of entries equal to two for $v_{i,c}$ are contained in the corresponding intervals for $v_{j,d}$. Hence $v_{i,c} \le v_{j,d}$ coordinatewise.

Conversely, assume $v_{i,c} \le v_{j,d}$. Comparing the first nonzero coordinate gives $i \ge j$. If $c \le r$ and $d < r$, then $c > d$ would give $v_{i,c}(c) = 1 > 0 = v_{j,d}(c)$; if $d \ge r$, then $c \le d$ is automatic. If $c > r$, the coefficient at α_r forces $d > r$. Were $c > d$, the terminal interval of entries equal to two for $v_{i,c}$ would begin strictly earlier than that for $v_{j,d}$, again contradicting coordinatewise domination. Thus $c \le d$, proving Equation (5).

Finally, the sum of the simple-root coefficients is $c - i + 1$ in both column regions, so root height minus one is $c - i$. $\square$

Corollary 3.2. *The poset P_r has ranks $0, \dots, H$, where $H = 2r - 2$, and has a unique maximum, represented by $(1, 2r-1)$.*

Proof. The minimum possible value of $c - i$ is zero. The largest is $(2r-1) - 1 = 2r - 2$, attained only at $(1, 2r-1)$. $\square$

4. Adjoining a minimum

The next argument is valid for any finite graded poset with a unique maximum.

Lemma 4.1. *Let P have ranks $0, \dots, H$ and a unique maximum. Adjoin a new minimum $\widehat{0}$ and denote the resulting bounded poset by $\widehat{P}$. Give it the shifted rank*

$$\widehat{\rho}(\widehat{0}) = 0, \qquad \widehat{\rho}(x) = \rho(x) + 1 \quad (x \in P).$$

Then

$$\mathcal{F}_{\widehat{P},\widehat{\rho}}(q, t) = \frac{\mathcal{F}_{P,\rho}(q, qt)}{1 - t}, \tag{7}$$

$$\mathbb{H}_{\widehat{P},\widehat{\rho}}(q, t) = \mathbb{H}_{P,\rho}(q, qt). \tag{8}$$

Proof. A weak chain of total length m in $\widehat{P}$ has a unique initial block of $m-\ell$ copies of $\widehat{0}$, followed by a weak chain of length ℓ in P. Shifting every old rank by one multiplies the weight of that suffix by q^ℓ. Therefore

$$A_m(\widehat{P},\widehat{\rho};q) = \sum_{\ell=0}^{m} q^\ell A_\ell(P,\rho;q).$$

Summing in m gives Equation (7).

Using Equation (1), the denominator on the right-hand side of Equation (7) is

$$(1-t)\prod_{a=0}^{H}(1-q^a(qt)) = \prod_{j=0}^{H+1}(1-q^j t),$$

which is exactly the prescribed denominator for $\widehat{\rho}$. Both sides have now been written over the same fixed denominator, so comparison of numerators proves Equation (8); no reduced-fraction assumption is required. □

Proposition 4.2. *With the hypotheses of Lemma 4.1,*

$$\mathbb{H}_{P,\rho}(q,t) = \sum_{S\subseteq[H]} \beta_{\widehat{P}}(S) q^{\sigma(S)-|S|} t^{|S|}. \tag{9}$$

In particular, every exponent of q on the right is nonnegative.

Proof. Apply Theorem 2.2 to $\widehat{P}$, then use Equation (8) and replace t by t/q. Since every $s \in S$ satisfies $s \geq 1$,

$$\sigma(S) - |S| = \sum_{s\in S}(s-1) \geq 0.$$

□

5. The augmented type-B lattice

Let

$$\widehat{L}_r = C_r \sqcup \{\widehat{0}\}, \qquad \widehat{\rho}(\widehat{0}) = 0, \qquad \widehat{\rho}(i,c) = c-i+1.$$

The new element $\widehat{0}$ lies below every cell.

Proposition 5.1. *The poset $\widehat{L}_r$ is a bounded graded lattice. For nonbottom cells $x=(i,c)$ and $y=(j,d)$,*

$$x \vee y = (\min(i,j), \max(c,d)), \tag{10}$$

$$x \wedge y = \begin{cases} (\max(i,j), \min(c,d)), & \max(i,j) \leq \min(c,d), \\ \widehat{0}, & \max(i,j) > \min(c,d). \end{cases} \tag{11}$$

Its rank function is $\widehat{\rho}$.

Proof. Put $I = \min(i,j)$ and $C = \max(c,d)$. We have $I \leq C$, and

$$C \leq \max(2r-i, 2r-j) = 2r - I,$$

so the cell in Equation (10) always lies in C_r. The coordinate order in Equation (5) shows that it is the least common upper bound.

For the proposed nonbottom meet, put $J = \max(i,j)$ and $D = \min(c,d)$. When $J \leq D$,

$$D \leq \min(2r-i, 2r-j) = 2r - J,$$

so (J,D) is a cell and is the greatest common lower bound. When $J > D$, any nonbottom common lower bound would need first coordinate at least J and second coordinate at most D, contradicting the defining cell inequality that the first coordinate is at most the second. Its meet is therefore $\widehat{0}$.

Among cells, a cover changes exactly one of $-i$ and c by one while remaining in C_r; the new minimum is covered precisely by the diagonal cells (i,i). Equivalently, if two comparable cells have rank difference at least two, changing i downward by one when possible, or increasing c by one otherwise, supplies an intermediate cell. Thus $\widehat{\rho}$ is the lattice rank. $\square$

Proposition 5.2. *The lattice $\widehat{L}_r$ is lower semimodular. Equivalently, its rank is supermodular:*

$$\widehat{\rho}(x) + \widehat{\rho}(y) \leq \widehat{\rho}(x \wedge y) + \widehat{\rho}(x \vee y) \tag{12}$$

for all $x, y \in \widehat{L}_r$.

Proof. Pairs involving $\widehat{0}$ give equality. For two cells with nonbottom meet, direct substitution of Equations (10) and (11) also gives equality.

Suppose the meet is $\widehat{0}$. Interchanging x and y if necessary, assume $i \geq j$. The failed cell condition is $i > \min(c,d)$. Since $c \geq i$, it follows that $d < i \leq c$. Consequently $x \vee y = (j,c)$, and

$$\widehat{\rho}(x) + \widehat{\rho}(y) - \widehat{\rho}(x \vee y) = d - i + 1 \leq 0.$$

Because $\widehat{\rho}(x \wedge y) = 0$, this is Equation (12). $\square$

Independently of the preceding structural property, the cell coordinates give the direct labelling needed for the descent formula.

Proposition 5.3. *Label the covers of $\widehat{L}_r$ by integers, with their usual order, as follows:*

$$\begin{aligned} \lambda(\widehat{0},(i,i)) &= i, \\ \lambda((i,c),(i-1,c)) &= -(i-1), \\ \lambda((i,c),(i,c+1)) &= c+1. \end{aligned} \tag{13}$$

Whenever the displayed pairs are covers, this is an EL-labelling of $\widehat{L}_r$. In particular the labels along any maximal chain are pairwise distinct, so the strictly increasing chain is also the unique weakly increasing chain.

Proof. Consider first an interval from (i,c) to (j,d), with neither endpoint equal to $\widehat{0}$. Every maximal chain is a shuffle of the left-expansion labels

$$-(i-1), -(i-2), \ldots, -j$$

and the right-expansion labels

$$c+1, c+2, \ldots, d,$$

with each displayed list used in its given order. Both lists are strictly increasing. Because every left label is negative and every right label is positive, the unique increasing shuffle performs all left expansions before all right expansions. At every step where both moves are available, it chooses the smaller label, so it is also lexicographically first.

Now consider an interval from $\widehat{0}$ to (j,d). Its unique increasing maximal chain begins with the atom (j,j), then makes only right expansions; its labels are

$$j, j+1, \ldots, d.$$

A chain beginning at an atom (i,i) with $i > j$ must eventually make a left expansion, producing a positive-to-negative descent. Moreover, the initial label j is the smallest label of an atom below (j,d). Hence the displayed chain is uniquely increasing and lexicographically first. Degenerate intervals are unique chains, so these cases cover every interval. $\square$

6. EL descents and the ballot-word formula

We now identify the maximal chains and descents of the explicit labelling in Proposition 5.3. A cover after the initial atom is called an L-move if it decreases the first cell coordinate and an R-move if it increases the second.

Lemma 6.1. *Reading the L- and R-moves gives a bijection from the maximal chains of $\widehat{L}_r$ to $\mathcal{B}_r$. Under this bijection, the descent set of the label sequence is $D(w)$.*

Proof. A maximal chain begins at an atom (i,i) and ends at $(1,2r-1)$. After the initial edge, it has $i-1$ left moves and $2r-1-i$ right moves, hence $m=2r-2$ moves in total. If L_k and R_k count the two moves in the first k letters, the intermediate cell is

$$(i-L_k, i+R_k).$$

Its right boundary condition is

$$R_k - L_k \le 2(r-i). \tag{14}$$

The remaining cell inequalities $1 \le i-L_k \le r$ and $i-L_k \le i+R_k$ hold automatically: there are only $i-1$ letters L in the whole word, so the first coordinate decreases from i to 1 without leaving $\{1,\dots,r\}$. At the end equality holds in Equation (14). Consequently, Equation (14) for all prefixes is equivalent to saying that every suffix of the word has nonnegative R-minus-L balance. This is exactly the ballot-prefix condition on the reversed word.

Conversely, let $w \in \mathcal{B}_r$, and put $b = \#R - \#L$. The ballot condition gives $0 \le b \le 2r-2$, and b is even. Thus $i = r - b/2$ is an integer in $[1,r]$. Starting at (i,i) and following w, the reversed ballot inequalities are exactly Equation (14); the path therefore stays in C_r and terminates at $(1,2r-1)$. This constructs the inverse bijection.

It remains to read the descents. The maximal chain has $2r-1$ covers, and descents occur at internal ranks $1,\dots,2r-2$. The initial label is the positive integer i. Consecutive left moves have increasing negative labels, consecutive right moves have increasing positive labels, and a left move followed by a right move is also increasing. Comparing the initial label with the first move therefore produces a descent at rank 1 if and only if that move is left. For $j \ge 2$, comparing the $(j-1)$-st and j-th moves produces a descent at rank j if and only if a right move is followed by a left move. These are exactly the elements of $D(w)$. ☐

Proof of Theorem 1.1. By Lemma 3.1 and Proposition 5.1, adjoining a minimum to P_r gives the bounded graded lattice $\widehat{L}_r$. The explicit EL-labelling of Proposition 5.3 is an R-labelling, so Equation (4) and Proposition 4.2 give

$$\mathbb{H}_{P_r,\mathrm{rk}}(q,t) = \sum_{\mathfrak{c}} q^{\sigma(\mathrm{Des}(\mathfrak{c})) - |\mathrm{Des}(\mathfrak{c})|} t^{|\mathrm{Des}(\mathfrak{c})|},$$

where the sum is over the maximal chains of $\widehat{L}_r$. Apply Lemma 6.1: the chains become the words $w \in \mathcal{B}_r$, their descent sets become $D(w)$, and

$$\sigma(D(w)) - |D(w)| = \sum_{j \in D(w)} (j-1).$$

This is the asserted formula. Every summand is a monomial with coefficient one and nonnegative exponents, proving coefficientwise nonnegativity as well. For $r=1$, the unique chain corresponds to the empty word and the formula gives $\mathbb{H}_{P_1,\mathrm{rk}}(q,t) = 1$. ☐

The formula gives sharp global information that is not visible from nonnegativity alone.

Corollary 6.2. *For every $r \ge 1$,*

$$\mathbb{H}_{P_r,\mathrm{rk}}(1,1) = \binom{2r-2}{r-1}, \qquad \deg_t \mathbb{H}_{P_r,\mathrm{rk}} = r-1,$$

and the coefficient of the highest power of t is

$$[t^{r-1}]\mathbb{H}_{P_r,\mathrm{rk}}(q,t) = q^{(r-1)(r-2)}.$$

Moreover, for $0 \le k \le r-1$,

$$[t^k]\mathbb{H}_{P_r,\mathrm{rk}}(1,t) = \binom{r-1}{k}^2.$$

Lemma 6.3. *Let $a(n,k)$ be the number of words of length $2n$ whose every prefix has at least as many R's as L's and which have exactly k L-runs. Then*

$$a(n,k) = \binom{n}{k}^2.$$

Proof. Let $C(z,u)$ be the generating function of Dyck words, with z marking semilength and u marking L-runs, using the same convention that R is an up-step. The first-return decomposition gives

$$C = 1 + zC(C+u-1).$$

Indeed, in a first-return factor $RALB$, the outer pair creates a new L-run exactly when A is empty. (The resulting generating function is the classical Narayana generating function for Dyck paths by number of peaks; see [6].) Every ballot word has a unique factorization

$$D_0RD_1R\cdots RD_h,$$

where the D_i are Dyck words and h is the final height. Taking the even-length part (so $h=2j$) therefore gives the generating function

$$E(z,u) = \sum_{n,k} a(n,k)z^nu^k = \frac{C(z,u)}{1-zC(z,u)^2}.$$

Put $Y = C-1$, so $Y = z(1+Y)(u+Y)$. The coefficients of E are extracted from this functional equation by Lagrange–Bürmann inversion. First expand

$$E(z,u) = \sum_{j\ge 0} z^jC(z,u)^{2j+1}.$$

For $s\ge 1$, $p\ge 1$, and $1\le k\le s$, inversion applied to $Y = z(1+Y)(u+Y)$ gives

$$\begin{aligned}[z^su^k]C(z,u)^p &= \frac{p}{s}[y^{s-1}u^k](1+y)^{p+s-1}(u+y)^s \\ &= \frac{p}{s}\binom{s}{k}\binom{p+s-1}{k-1}.\end{aligned}$$

Consequently, for $n\ge 1$ and $1\le k\le n$, with $s = n-j$,

$$[z^nu^k]E = \sum_{s=k}^{n}\frac{2n-2s+1}{s}\binom{s}{k}\binom{2n-s}{k-1}.$$

To evaluate this finite sum, use $\binom{s}{k}/s = \binom{s-1}{k-1}/k$, put $A = s-1$ and $B = 2n-1-A$, and apply Pascal's identity:

$$\begin{aligned}&\frac{B-A}{k}\binom{A}{k-1}\binom{B}{k-1} \\ &= \binom{A}{k-1}\binom{B}{k} - \binom{A}{k}\binom{B}{k-1} \\ &= \binom{A+1}{k}\binom{2n-A-1}{k} - \binom{A}{k}\binom{2n-A}{k}.\end{aligned}$$

The sum therefore telescopes from $A = k-1$ to $A = n-1$, and is $\binom{n}{k}^2$. For $k = 0$, setting $u = 0$ gives $C(z,0) = 1$ and $E(z,0) = 1/(1-z)$, so the coefficient is 1; for $n = 0$ the empty word gives the same conclusion. Hence $[z^n u^k]E(z,u) = \binom{n}{k}^2$ in all cases. □

Proof of Corollary 6.2. Write $n = r-1$. Reversal identifies $\mathcal{B}_r$ with the ballot words of length $2n$. By the reflection principle, the number ending at height $2k$ is

$$\binom{2n}{n-k} - \binom{2n}{n-k-1} \qquad (0 \le k \le n),$$

where $\binom{2n}{-1} = 0$. Summing over k telescopes to $\binom{2n}{n}$, proving the first identity.

Every reversed ballot word has at most n letters L, so $|D(w)| \le n$. Equality forces exactly n letters L, each starting a separate run. For $n = 0$ the unique word is empty. For $n \ge 1$, a nonempty word in $\mathcal{B}_r$ ends in R, and the unique such word with n isolated letters L is

$$w = (LR)^n.$$

Its descent positions are $1, 3, \dots, 2n-1$, and hence its q-exponent is

$$\sum_{a=0}^{n-1} 2a = n(n-1).$$

This proves both the sharp degree and the stated leading coefficient.

It remains to evaluate the coefficients at $q = 1$. By Theorem 1.1, the coefficient of t^k is the number of ballot words of length $2n$ with exactly k runs of the letter L; reversal preserves the number of runs. Lemma 6.3 therefore yields

$$[t^k]\mathbb{H}_{P_r,\mathrm{rk}}(1,t) = \binom{r-1}{k}^2.$$

□

For completeness, the flag interpretation also yields a three-rank topological refinement. Recall that $H = 2r-2$. For $0 \le a < b < c \le H-1$, let L_a be rank a of P_r, set $f_a = |L_a|$, let e_{ab} count comparable pairs in $L_a \times L_b$, and let n_{abc} count chains $x < y < z$ through the three ranks.

Corollary 6.4. *Define*

$$\beta_{abc} = n_{abc} - e_{ab} - e_{ac} - e_{bc} + f_a + f_b + f_c - 1.$$

Then

$$[t^3]\mathbb{H}_{P_r,\mathrm{rk}}(q,t) = \sum_{0\le a<b<c\le H-1} \beta_{abc} q^{a+b+c}, \qquad \beta_{abc} \ge 0.$$

Moreover, β_{abc} is the reduced Euler characteristic of the order complex induced on $L_a \cup L_b \cup L_c$.

Proof. Apply Boolean inversion in Equation (2) to the set $\{a+1, b+1, c+1\}$, then use Equation (9). The induced three-rank order complex has $f_a+f_b+f_c$ vertices, $e_{ab}+e_{ac}+e_{bc}$ edges, and n_{abc} triangles, giving the Euler-characteristic formula. Nonnegativity follows from the descent interpretation in Equation (4). The Euler interpretation alone would not determine the sign. □

7. Further remarks

7.1. **Position in the literature.** The order-theoretic structure of positive roots, including their Hasse diagrams and antichain symmetries, has been studied extensively; a useful general reference is Panyushev [4]. Generalized Catalan enumeration for Weyl groups and root-poset antichains is developed in [1]. The statistic considered here is different: it weights weak chains by the sum of their root-poset ranks.

Chapoton introduced this q-Zeta invariant, established the bounded flag expansion, and observed that the type-B numerator appeared to have nonnegative coefficients which, as polynomials in q, specialise at $q = 1$ to squares of binomial coefficients [3]. Under the Lie-rank convention of this paper, the proved specialization is $[t^k]\mathbb{H}_{P_r,\mathrm{rk}}(1,t) = \binom{r-1}{k}^2$. The descent formula used above goes back to Stanley's work on Jordan–Hölder sets [5]. Björner's EL-labelling framework packages intervalwise increasing chains together with lexicographic shellings [2]. In the present setting, the signed endpoint labelling makes the relevant descent enumerator completely explicit and leads to the ballot-word formula in Theorem 1.1.

7.2. **Finite verification.** As an independent exact-arithmetic check, we generated the three families of positive roots of B_r in simple-root coordinates, matched them bijectively to C_r, and verified the coordinate order, the join and meet formulae, and the lower-semimodular rank inequality for $1 \le r \le 10$. The same check enumerates every interval of $\widehat{L}_r$ to verify the EL property, computes the q-Zeta numerator directly from the defining weak-chain sums, and matches it against the ballot-word formula, again for $1 \le r \le 10$. For these ranks the word counts agree with $\binom{2r-2}{r-1}$, the sharp top t-coefficient agrees with Corollary 6.2, and the coefficients of t^k at $q = 1$ agree with $\binom{r-1}{k}^2$. The small-rank polynomials displayed in Section 1 were checked separately.

These computations are exact finite reproduction for the named ranks only. They neither prove Theorem 1.1 nor justify extrapolation to higher rank. The all-rank formula rests on the proofs in Sections 4 to 6.

7.3. **Scope.** The result concerns the full positive-root poset of standard Lie type B_r, with r equal to the Lie rank, and Chapoton's fixed denominator normalization. The argument neither characterizes all posets with nonnegative q-Zeta numerator nor settles the corresponding questions for other Coxeter types or denominator conventions.

8. Conclusion

The q-Zeta numerator of the type-B_r positive-root poset is the descent enumerator of reversed-ballot words, with the q-weight recording the shifted positions of the left-run starts. This description arises from an exact minimum-adjunction identity and a signed EL-labelling of the trapezoidal cell lattice. It proves coefficientwise positivity in every Lie rank and, beyond the sign, determines the evaluation $\sum_k \binom{r-1}{k}^2 t^k$ at $q = 1$, the sharp t-degree, and the leading t-coefficient.

Acknowledgements

Qihang Wang thanks his doctoral advisor, Weinan E, for his encouragement and support. We also acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

Disclosure of automated assistance

OpenAI GPT-5.6 Sol was used during proof exploration and drafting. Anthropic Claude Fable 5, operating in an agentic workflow, was used for adversarial checking, literature comparison, and language editing. The argument in this paper is self-contained, no model output is used as a mathematical premise, and the authors assume full responsibility for the final manuscript.

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